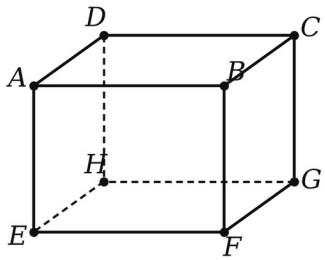
Honors Geometry – Ch. 3 Test PRACTICE **WORKED KEY**

Parallel and Perpendicular Lines • Sections 3.1 – 3.5

Show all work in the space provided. Check your work. Write each final answer on the line. Diagrams are not drawn to scale.

Use the box for questions 1 – 3.



1. Name a line parallel to line DH. (1)

→ AE, BF, or CG

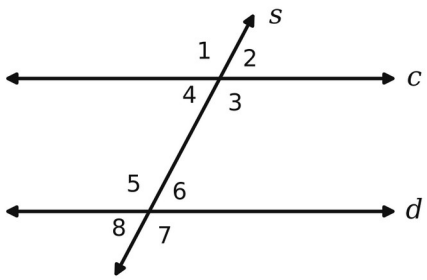
2. Name a line skew to line AD. (1)

→ BF, CG, EF, or GH

3. Name a plane parallel to plane ABF. (1)

→ plane DCG (DCGH)

Use the diagram for questions 4 – 9. In 7 – 9, c ∥ d.



4. Classify the angle pair: ∠1 and ∠5. (1)

→ corresponding

5. Classify the angle pair: ∠6 and ∠4. (1)

→ alternate interior

6. Classify the angle pair: ∠3 and ∠6. (1)

→ consecutive interior

7. m∠7 = 118°. Find m∠1. (2)

→ 118° (alternate exterior angles are congruent)

8. m∠3 = 121°. Find m∠6. (2)

→ 59° (consecutive interior angles are supplementary: 180 − 121)

9. m∠5 = 116°. Find m∠2. (2)

→ 64° (∠5 ≅ ∠1 (corresponding) = 116°; ∠1 and ∠2 are a linear pair: 180 − 116)

10. Two parallel lines are cut by a transversal. A pair of corresponding angles measure (5x − 7)° and 68°. Find x. (2)

→ x = 15

corresponding → EQUAL: (5x − 7) = 68 → x = 15

11. Two parallel lines are cut by a transversal. A pair of consecutive interior angles measure (4x + 6)° and (3x + 13)°. Find x and BOTH angle measures. (2)

→ x = 23; 98° and 82°

consecutive interior → ADD to 180: 7x + 19 = 180 → x = 23; 98° and 82°

Honors Geometry – Ch. 3 Test PRACTICE **WORKED KEY**

12. Use the diagram for 4 – 9, but suppose you do NOT know that c ∥ d. If ∠4 and ∠5 are supplementary, then c ∥ d. Name the theorem that proves it. (2)

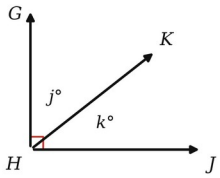
→ Consecutive Interior Angles Converse

13. Two lines are cut by a transversal. A pair of alternate interior angles measure (7x − 4)° and (4x + 20)°. Find the value of x that makes the lines parallel. (2)

→ x = 8

converse needs them CONGRUENT: 7x − 4 = 4x + 20 → x = 8 (both 52°)

14. In the diagram, GH ⊥ HJ. j = (2x + 16) and k = (4x − 10). Find x. (2)



→ x = 14

⊥ → ∠GHJ = 90°, so j + k = 90: 6x + 6 = 90 → x = 14 (j = 44°, k = 46°)

15. In a plane, line s ⊥ line u and line t ⊥ line u. What can you conclude about lines s and t? Name the theorem. (2)

→ s ∥ t; Lines Perpendicular to a Transversal Theorem

16. In a plane, line c ∥ line d and line e ⊥ line c. What can you conclude about lines e and d? Name the theorem. (2)

→ e ⊥ d; Perpendicular Transversal Theorem

17. Point P lies on the perpendicular bisector of AB. PA = 4x + 1 and PB = 6x − 9. Find PA. (2)

→ PA = 21

on the ⊥ bisector → EQUIDISTANT: 4x + 1 = 6x − 9 → x = 5; PA = 21

18. Find the slope of the line through (−2, 7) and (6, 1). (1)

→ m = −3/4

m = (1 − 7) / (6 − (−2)) = −6/8 = −3/4

19. Parallel, perpendicular, coinciding, or neither? Explain. $y = \frac{1}{3}x + 4$ and $x - 3y = 9$ (2)

→ parallel

x − 3y = 9 → y = ⅓x − 3: same slope, different y-intercepts

20. Write an equation in slope-intercept form of the line through (3, −1) that is PARALLEL to y = 4x − 2. (2)

→ y = 4x − 13

same slope 4: −1 = 4(3) + b → b = −13

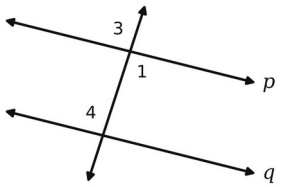
21. Write an equation in slope-intercept form of the line through (2, 7) that is PERPENDICULAR to x − 4y = 8. (2)

→ y = −4x + 15

x − 4y = 8 → y = ¼x − 2, slope ¼; perpendicular slope −4; 7 = −4(2) + b → b = 15

22. Complete the proof. Given: p ∥ q. Prove: ∠1 ≅ ∠4. (5)

Do NOT use the Alternate Interior Angles Theorem.



Statements	Reasons
1. p ∥ q	**1.** Given
2. ∠3 ≅ ∠4	**2.** Corresponding Angles Postulate
3. ∠1 ≅ ∠3	**3.** Vertical Angles Theorem
4. ∠1 ≅ ∠4	**4.** Transitive Property of Congruence